

# Fei Yi

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## EDUCATION

### Emory University

Atlanta, GA

B.S. in Quantitative Science - Data Science | Minor in Artificial Intelligence

Graduated May 2026

- Cumulative GPA: 3.82/4.0
- Relevant Courses: Machine Learning, Quantitative Models, Probability, Linear Algebra, Regression Analysis
- Technical Skills: Python, R, SQL, JavaScript, pandas, NumPy, scikit-learn, PyTorch, NLP, time-series forecasting, Matplotlib, Seaborn, Excel, Google Analytics, Git, Jupyter, VS Code, ChatGPT API, AI-assisted development / vibe coding

## WORK EXPERIENCE

### Tencent Cloud

Shenzhen, China

Data Science Intern

Jun. 2024 - Aug. 2024

- Built an end-to-end client demand forecasting pipeline with Python, SQL, and Tencent's Hunyuan AI framework to surface forward-looking sales signals.
- Aggregated client usage logs from multiple internal sources, cleaned inconsistent IDs and missing timestamps, and engineered rolling, trend, and seasonality features.
- Designed a self-service sales decision dashboard that translated forecasts into client-level risk, growth, and outreach priorities.
- Optimized data extraction workflows, reducing processing time by ~15% and enabling faster refresh cycles for decision-making.

### Atmosfy, Inc.

San Francisco, CA

Data Analysis Intern (Remote)

Sep. 2024 - Dec. 2024

- Analyzed market, local news, and user behavior datasets to support weekly growth research and product decision-making.
- Drafted weekly 10-page analytical memos highlighting market developments, engagement patterns, and retention opportunities.
- Collaborated on a user churn prediction workflow using SQL and Python, translating behavioral signals into targeted retention insights.

### Emory University Quantitative Science Department | Prof. Clifford J. Carrubba

Atlanta, GA

Research Assistant

Jan. 2025 - May 2026

- Studied dynamic strategic interaction models with evolutionary game theory to examine cooperation and defection among AI agents.
- Scraped operational data from 50+ global AI models to construct a panel dataset for fixed-effects validation of game-theoretic predictions.
- Analyzed computational centralization dynamics and identified conditions where competitive agent behavior reduced system-wide sustainability.

## PROJECT AND RESEARCH EXPERIENCE

### GridStateFormer: Next-Day Electricity Demand Forecasting

Atlanta, GA

Machine Learning Researcher

Spring 2026

- Developed a hybrid PatchTST and state-space model for 24-hour regional electricity demand forecasting using U.S. EIA hourly load data across 148 regions.
- Built a 168-hour input to 24-hour forecast pipeline with RevIN normalization, daily patch encoding, latent rollout, and seasonal residual forecasting.
- Benchmarked against GRU, LSTM, TCN, CNN, and seasonal-naive baselines; achieved RMSE 0.3903, MAE 0.2381, and R2 0.8131.

### Online Discourse by Federal Officials on ICE and Immigration

Atlanta, GA

Data Engineering & Political Time-Series Analysis Lead

Spring 2026

- Scraped and processed large-scale political social media data covering 1.57M posts from 541 federal actors, including ~80K ICE/immigration-related posts.
- Led Error Correction Model (ECM) analysis to separate short-term attention shocks from long-run equilibrium patterns in political discourse.
- Modeled how event-driven spikes, actor participation, and platform dynamics shaped ICE-related attention during January 2025 - March 2026.

### Drug Side Effects - NLP Clustering Analysis

Atlanta, GA

Unsupervised Learning Researcher

Spring 2025

- Clustered 2,900+ drugs using side-effect text representations, showing that adverse-effect profiles captured pharmacological similarity better than names or conditions.
- Compared MinHash LSH, TF-IDF, DBSCAN, and agglomerative clustering, demonstrating that feature representation drove clustering quality more than algorithm choice.

## ADDITIONAL INFORMATION

Language: Mandarin Chinese (native), English (full proficiency), German (intermediate), Japanese (basic)

Volunteering: Michael C. Carlos Museum Volunteer, Hainan Youth Supportive Teacher

Interests: Tennis, Video-gaming, Hiking, Photography, Cuisine, Movies, Vlogging and Video Editing